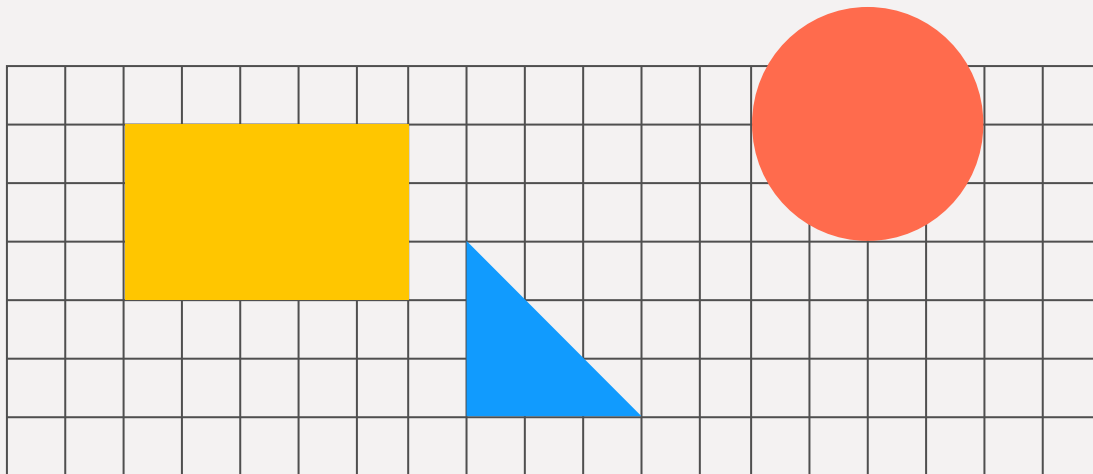


Teacher Education and Certification Requirements in Charter Schools and Their Impact on Student Achievement

Kathryn, Anika, Rachel, and Kira



Research Question and Motivation

Research Question: How do relaxed requirements for charter schools (represented by looser teacher licensure requirements) impact North Carolina's 2013 policy lowering the required share of licensed teachers in charter schools from 75% to 50% affect teacher licensure rates in elementary charter schools? After the policy took effect in 2014, how did this relaxation of charter-specific teacher licensure requirements impact student achievement, measured by the percentage of elementary students passing standardized assessments?

Motivation:

- Charter school enrollment more than doubled between fall 2010 and fall 2021, reaching 3.7 million students. (NCES 2023)
- Charter schools are now a major part of the public education system, and the policies governing them have large-scale impacts.
- Charter schools have long been scrutinized for having more relaxed requirements for their staff, sparking controversy over their quality of education. Counterarguments to this include allowing for more subject-specific, non-teacher certified educators into classrooms (Hess 2018).

Why this Matters: From a policy perspective, teacher licensing requirements sit at the core of the charter school model's central tradeoff: flexibility versus accountability.

Debate:

- Proponents argue that easing certification requirements allows schools to recruit individuals with strong subject-matter expertise or industry experience, potentially improving instruction, especially in hard-to-staff subjects (Darling-Hammond 2002).
- Critics, on the other hand, argue that formal training and certification are essential for effective teaching, and that reducing these requirements may harm students (Darling-Hammond 2002).

Economic Analysis:

- Human capital theory gives us a lens to evaluate whether teacher credentials function as genuine skill signals or merely as costly barriers to entry, and whether the investment in licensure produces measurable returns in student learning.
- Empirical methods will allow us to isolate the causal impact of different regulatory regimes on student performance.

Charter Schools, Loose Hiring Practices, and Student Achievement

A large body of literature examines the impact of charter schools on student achievement

- Overall, the effects are inconsistent, likely due to heterogenous effects that occur as a result of the varying regulations and organizational requirements that change by state (Dallavis and Berends, 2023).
 - Geographical area plays a key role in determining achievement effects, with studies conducted in urban areas finding larger positive effects than studies focusing rural or diverse geographical areas (Betts and Tang, 2011).
 - Similarly, grade levels for charter schools also play a role in determining student achievement effects. Elementary and middle school reading and math scores see no negative effect in charter schools, compared to high schools that typically view negative effects when compared to public schools (Betts and Tang, 2011).

One aspect influencing student achievement in charter schools is different regulations regarding teacher hiring practices when compared to public schools. Charter schools have more flexibility regarding who they can hire to teach (Edwards & Hall, 2018).

- Charter school administrators *are* taking advantage of opportunities to be innovative in their hiring practices and teachers in charter schools look different than their colleagues in traditional public schools. It is not entirely clear whether this is a positive development since charter school teachers look stronger on some quality measures and weaker on others (Baker and Dickerson, 2019).
 - As a result of these hiring practices and policies, charter schools tend to have less experienced teachers and higher teacher turnover (Carruthers, 2023).

Charter Schools, Loose Hiring Practices, and Student Achievement

Many charter school hiring practices also differ state by state. One of the main regulation areas that differs is teacher licensures. Teacher licensure requirements have become heavily debated as the supply for teachers continues to diminish, partially due to high barriers to entry (Chi et al., 2023).

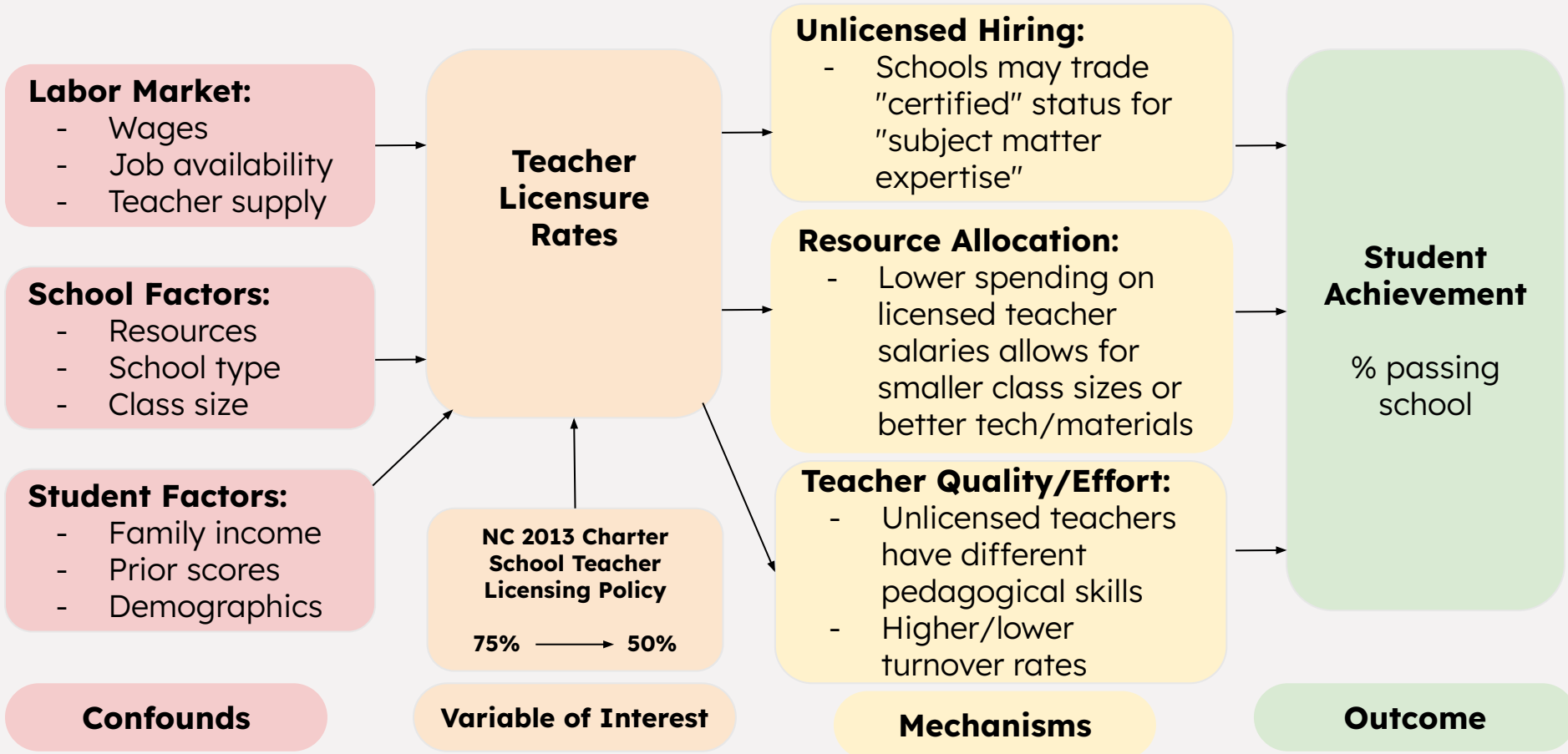
- Studies focusing on high school teacher licensures point to a positive relationship between strict licensure requirements and student achievement in charter schools. The actual size of these gains is mixed (Darling-Hammond, 2000)
- On the other hand, studies that focus on K-12 teacher licensure regulations find no relationship between a teacher licensure requirement and student achievement (Buddin and Zamarro, 2008)
- Many teachers report that licensure requirements provide clarity on their teaching requirements and improve education quality (Ghamwari, 2023).

Most studies regarding teacher hiring regulations rely on panel and longitudinal data and, as a result, find primarily correlational or quasi-casual relationships between the unique hiring practices and student achievement (Epple et al., 2015).

- Barrett et al. (2021), Clotfelter et al. (2007), and Harris and Sass (2011) employ longitudinal panel data, controlling for school fixed effects to study the value added of teacher training and licensure requirements.
 - They were able to claim a quasi-casual relationship between that and student achievement.

Our study adds to the debate on Charter schools regulation, student achievement, and teacher licensure by examining how changes in North Carolina charter teacher licensure requirements may affect hiring practices and outcomes in elementary and middle schools. It is distinctive because it uses a difference-in-differences analysis of a real policy change, which is an uncommon approach to this topic of study

Conceptual Model



Policy Background:

What Qualifies a Licensed Teacher

Obtaining North Carolina Teacher Licensure requires a bachelor's degree, completion of a state-approved Educator Preparation Program (EPP), and passing required Pearson or Praxis content exams (North Carolina Department of Public Instruction).

SESSION LAW 2013-355 | SENATE BILL 337 ~ "§ 115C-238.29F. General requirements.

(e) Employees. –

(1) An employee of a charter school is not an employee of the local school administrative unit in which the charter school is located. The charter school's board of directors shall employ and contract with necessary teachers to perform the particular service for which they are employed in the school; at least ~~seventy-five~~fifty percent (~~75%~~50%) of these teachers in grades kindergarten through five, at least fifty percent (50%) of these teachers in grades six through eight, and at least fifty percent (50%) of these teachers in grades ~~nine through 12~~ shall hold teacher certificates-licenses. All teachers in grades six through 12 who are teaching in the core subject areas of mathematics, science, social studies, and language arts shall be college graduates.

The board also may employ necessary employees who are not required to hold teacher certificates-licenses to perform duties other than teaching and may contract for other services. The board may discharge teachers and ~~noncertificated~~ nonlicensed employees.

- In 2013, North Carolina Senate passed Bill 337, “An act to create the North Carolina Charter Schools advisory board and make other changes to charter school laws.” In clause 238.29F, a policy reform to relax the percentage of licensed teachers per charter school from 75% to 50% was passed (*S.L. 2013-355*, 2017).
- Governor Pat McCrory advocated for this change in order to increase flexibility, specifically surrounding elementary school hires. The figure of “50%” itself was highly contested until the last session, as it originally was going to be even further relaxed. The main argument was that licensure and teaching quality are not correlated; this reform would theoretically allow charter schools to pilot non-licensed teachers in the state (North Carolina Cuts, 2013).

Data Sets Used

Data Source: NC School Report Card Data

The primary data source for this analysis is the North Carolina School Report Card (SRC) dataset, published annually by the North Carolina Department of Public Instruction (NC DPI). Report cards are provided for all district, charter, and alternative schools operating during each school year.

In most instances, data are reported at the school, district, and state levels, with school-level data based on information from all grades within the school.

The data are administrative in nature, collected through NC DPI's standard reporting systems rather than through surveys or experiments, which means coverage is near-universal but variable definitions and reporting practices may shift across years.

Question 1: Effect of 2013 Policy on Teacher Licensure

Uses Report Card Data (RCD) on:

- proportion of teacher licenses per charter school, filtered to remove high schools
- Which schools are public vs. charter

Question 2: Effect of Teacher Licensure on Student Achievement

Uses Report Card Data (RCD) on:

- proportion of teacher licenses per school, filtered to remove high schools
- Which schools are public vs. charter
- School-level achievement rate
- School-level demographic data

Key Variables

Variable	Used In	Role	How It Is Measured
Teacher licensure rate	Model 1 Model 2	Model 1: Outcome Model 2: Predictor	Share of teachers at the school who hold a full North Carolina teaching license, calculated each year from statewide license records. Ranges from 0 to 1.
Charter school status	Model 1	Treatment indicator	A dummy variable equal to 1 if the school is a charter school and 0 if it is a traditional public school. Does not vary over time for a given school.
Post-policy period (2014 onwards)	Model 1	Time indicator	A dummy variable equal to 1 for the 2014 academic year and later, the first full year after the 2013 law lowered the required licensed-teacher share for charter schools from 75% to 50%. Equal to 0 for years before 2014.
Student achievement rate	Model 2	Outcome	Percentage of students at the school who meet or exceed the grade-level proficiency standard on standardized assessments, aggregated across all tested grades and subjects for the full student population.
School demographic controls	Model 2	Controls	Annual school-level shares of Black students, white students, and economically disadvantaged students, drawn from enrollment records. Included to account for compositional differences across schools and years.
School level	Model 1	Filter	Categorical indicator of whether a school serves elementary, middle, or combined grades. High school-only campuses are excluded from the control group because their teacher workforce characteristics differ substantially.

Summary Statistics and Key Metrics

Table 1: Summary Statistics

	Mean	SD	Min	Max	N
% Above L3 (Math)	53.02	18.50	0.00	94.80	9862
% Fully Licensed	0.96	0.07	0.58	1.00	9862
% White	0.13	0.08	0.00	0.37	9862
% Black	0.07	0.06	0.00	0.33	9862
% Economically Disadvantaged	0.14	0.05	0.00	0.28	9862

Note:

Sample restricted to 2014–2018. N reflects estimation sample used in regression models. Outliers in %Full Licensed were removed so minimum reflects 25th percentile of original sample

Average Licensing Rates by Year

year	mean_licensed	n
2011	0.848	93
2012	0.807	94
2013	0.869	101
2014	0.850	117
2015	0.807	137
2016	0.776	149
2017	0.748	159
2018	0.811	166

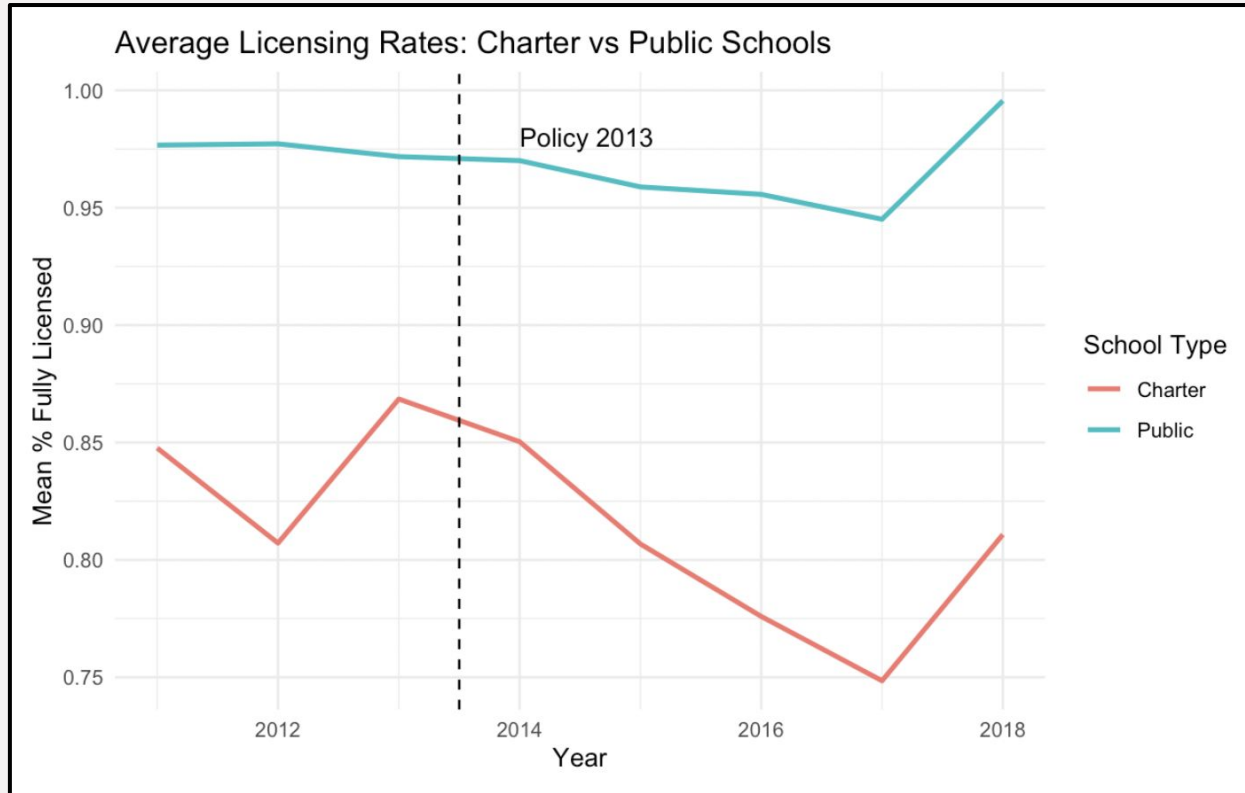
The post-2014 steady drop in charter licensing rates is consistent with the policy as schools hire more unlicensed teachers

The sudden spike in licensure in 2018 may be due to a relaxation of teacher licensing criteria, possibly correlating with increased ease of licensing

Key Metrics for Pre and Post Treatment and Control

group	period	mean_licensed	n_obs
Charter	post_2014	0.796	728
Charter	pre_2014	0.842	288
Public	post_2014	0.965	9729
Public	pre_2014	0.975	5774

Parallel Trends Limitation



A possible limitation for this study is that the pre-treatment data between public and charter schools does not indicate a clear parallel between the two. Charter schools were already licensing at lower rates relative to public schools (wherein all teachers must be licensed) pre 2013, meaning our data may indicate an acceleration of an existing dynamic but not the cause of it.

Furthermore, 2012's data may show a significant difference due to the cap for charter schools being lifted in NC in August of 2011. Up until this point, no more than 100 charter schools could exist in the state. Lifting the policy expanded development of more charter schools, likely leading to lower licensing rates in new charter school staff.

Empirical Model for the Effect of the Policy on Reducing Teacher Licensing

$$L_{st} = \alpha + \delta(C_s \times D_t) + \beta C_s + \gamma_s + \gamma_t + \epsilon_{st}$$

L_{st} = share of fully licensed teachers at school s in year t
 C_s = 1 if the school is charter, 0 if traditional public school

D_t = 1 if year was after 2014, 0 if year was before 2014

γ_t = year fixed effects

γ_s = school fixed effects

δ = research parameter of interest

δ is our parameter of interest and is causal because of the time and treatment control from the DiD model. School FE was not needed because βC_s controls for baseline differences and FE would absorb parameters of interest.

Possible Limitations:

- Concurrent policy changes
- New charter schools opening have a different composition

Empirical Model for Effect of the Percentage of Teachers Licensed on Student Achievement in Charter Schools

$$A_{st} = \alpha + \beta_1 L_{st} + \beta_2 L_{st}^2 + \mathbf{X}_{st} \delta + \gamma_t + \gamma_s + \epsilon_{st}$$

A_{st} = percent of students meeting an academic standard at school s in time t

L_{st} = share of fully licensed teachers at school s in year t

L_{st}^2 = quadratic term for nonlinear effects

$\mathbf{X}_{st}$ = time varying school controls: %black, %white, and %economically disadvantaged

γ_t = year fixed effects

γ_s = school fixed effects - to study within-school changes over time

$\beta_{1,2}$ = research parameters of interest, highlighting the licensing and achievement relationship

β is not causal, omitted variable bias from concurrent events remains in the model despite school FE. Parameters of interest are no significant with school FE.

Possible limitations:

- Other time varying confounders within schools like school funding or administration changes
- Demographic controls explain a majority of the variation (R^2 reduced from $\sim .6$ to $\sim .15$ when $\mathbf{X}_{st}$ is added)
- Reverse causality: lower achieving schools may hire more non-licensed teachers

2014 Policy Regression Results

Interpreting the coefficients

Model 1 - Regression Results:	
Impact of 2014 Policy on Charter School Licensing	
	(1)
Charter School	-0.133*** (0.013)
Charter × Post-2014	-0.036*** (0.011)
Num.Obs.	16495
R2	0.273
R2 Adj.	0.272
R2 Within	0.242
R2 Within Adj.	0.242
FE: year	X
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001	

- The Charter School **coefficient of –0.133** in the year FE model indicates that, even before the policy, charter schools had a **13.3 percentage point lower** share of licensed teachers than public schools, a meaningful pre-existing gap.
- The key DiD coefficient, Charter × Post-2014, equals –0.036 in the preferred model: after the policy took effect, charter schools reduced their licensed teacher share by an **additional 3.6 percentage points** relative to public schools.
- The 3.6 percentage point decline represents a **4.3% reduction relative to the pre-policy elementary charter school average of 84.2%**, indicating that while charter schools did respond to the loosened licensing requirement, the behavioral shift was modest in magnitude.

Student Pass Rates Regression Results

	Cross-Sectional OLS Models					Longitudinal FE
	Baseline	Diminishing Ret.	By Grade Level	By School Type	Full Specification	Within-School FE
Teacher Licensure Rate (%)	24.772*** (3.512)	12.405 (44.678)	36.000 (42.780)	94.451* (43.667)	9.360 (47.300)	46.638 (31.597)
Charter School (1=Yes)	0.525 (1.524)	0.507 (1.518)	-0.984 (1.478)	25.066** (8.974)	-0.706 (10.220)	
Licensure Rate Squared (Non-linear)		7.137 (25.159)	-48.868+ (25.588)	-35.921 (24.564)	-31.780 (26.455)	-26.312 (16.105)
% White Students	41.430*** (6.013)	41.389*** (6.010)	41.773*** (5.980)	41.504*** (5.971)	41.193*** (5.976)	63.729*** (11.305)
% Black Students	-44.565*** (7.285)	-44.521*** (7.278)	-43.038*** (7.196)	-43.654*** (7.231)	-42.379*** (7.197)	-81.868*** (13.990)
% Economically Disadvantaged	-163.172*** (5.695)	-163.176*** (5.691)	-162.687*** (5.601)	-163.011*** (5.653)	-162.427*** (5.595)	-14.379** (5.153)
Year (Centered)	0.516*** (0.075)	0.511*** (0.075)	0.526*** (0.075)	0.506*** (0.075)	0.510*** (0.075)	1.352*** (0.074)
Year Squared					-0.457*** (0.051)	
Licensure × Charter Status				-28.138** (10.030)	0.083 (11.514)	-11.163 (7.283)
Num.Obs.	9862	9862	9862	9862	9862	9816
R2	0.602	0.602	0.609	0.603	0.611	0.921

Note: Standard errors are clustered at the school level (agency_code). Demographic controls include race and socioeconomic status. Grade level interactions account for differences between elementary, middle, and high school licensing impacts.

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: This table is made only using math pass rates.

Interpreting the coefficients

- A 1 percentage point increase in the share of licensed teachers is associated with a **24.8 point increase** in pass rate (p < 0.001)
- Once school fixed effects are added the **estimate shifts to 46.6 (SE = 31.6) and loses statistical significance**, suggesting the baseline result partly reflects differences across charter schools rather than a within-school causal effect.
- The instability across specifications suggests caution in interpreting the licensure rate as a clean causal driver of achievement in charter schools.

Robustness of Models

Robustness - Model 1

Model 1 - Regression Results:
Robustness of 2014 Policy Impact

	Year FE	School & Year FE
Charter School	-0.133 (0.013)	
Charter × Post-2014	-0.036 (0.011)	-0.012 (0.010)
Count (N)	16495	16442
R ²	0.273	0.727

- When school and year fixed effects are added, the Charter × Post-2014 coefficient changes to −0.012 (SE = 0.010), which is no longer statistically significant.
- This change suggests that some of the estimated effect in the year FE model may reflect differences across schools that existed prior to the policy, not purely the policy's causal impact.

Robustness - Model 2

- Demographic controls are stable and large across all specifications, % economically disadvantaged ($\approx -163^{***}$) dominates, confirming student composition is the strongest predictor of pass rates.
- The licensure coefficient is sensitive to whether school FEs are included, which is the key robustness concern for this model.
- R^2 rises from 0.602 to 0.921 when moving to within-school FE, meaning most variation in charter school achievement is explained by stable school-level characteristics rather than licensure rates alone.

Student Demographics

Mechanism Analysis: Heterogeneous Effects				
	White Intensity	Black Intensity	Socioeconomic	Combined Mech.
Teacher Licensure Rate (%)	44.263 (35.372)	47.665 (37.845)	61.773+ (33.773)	59.131 (37.530)
Licensure Rate Squared	-24.662 (19.411)	-26.816 (18.982)	-31.332+ (16.650)	-24.910 (19.543)
Share of White Students	69.178* (30.633)	63.747*** (11.316)	63.983*** (11.318)	109.908* (48.982)
Share of Black Students	-81.720*** (14.029)	-80.125* (32.333)	-80.549*** (13.868)	-60.930 (47.395)
Share of EDS Students	-14.334** (5.161)	-14.367** (5.152)	29.872 (40.783)	50.148 (48.173)
Year (Centered)	1.352*** (0.074)	1.353*** (0.074)	1.345*** (0.073)	1.342*** (0.073)
Licensure × Charter	-10.413 (8.476)	-11.292 (7.723)	-14.771+ (7.941)	-11.412 (8.477)
Licensure × % White	-5.635 (28.591)			-47.193 (47.296)
Licensure × % Black		-1.872 (29.592)		-19.116 (45.263)
Licensure × % EDS			-47.141 (41.702)	-68.208 (49.341)
Num.Obs.	9816	9816	9816	9816
R2	0.921	0.921	0.921	0.921
FE: agency_code	X	X	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Mechanism #1: Teacher licensure affects student achievement differently depending on student demographics, schools serving more disadvantaged populations see the largest negative effects when licensure rates fall.

- The Licensure × EDS interaction (−47.1 to −68.2) suggests licensed teachers matter most for low-income students.
- The Licensure × Charter interaction is consistently negative across all columns (−10.4 to −14.8), meaning licensed teachers produce smaller achievement gains in charter schools than public schools.
- Share of Black students is a large negative predictor ($\approx -80^{***}$) but the Licensure × Black interaction is small and insignificant, licensure does not meaningfully close racial achievement gaps.
- Economically disadvantaged share drives the largest within-school penalty, confirming student socioeconomic composition is the dominant moderator of the licensure-achievement relationship.
- R^2 is 0.921 across all columns with school fixed effects, meaning all findings reflect within-school changes over time rather than differences across schools.
- Results align with the Teacher Quality/Effort channel in the conceptual model, licensed teachers matter most for vulnerable students, and charter schools see a weaker licensure-achievement link consistent with trading credentials for subject expertise.

Subject-Specific Turnover Rates

Mechanism #2: The 2014 policy lowered licensure requirements, which drove different rates of teacher turnover across subjects, and these subject-specific turnover patterns are reflected in the differential impacts on student achievement across math, reading, and science.

- The study finds that a traditionally certified science teacher had a statistically significant positive effect on Biology end-of-course STAAR scores, while a teacher certified through an online alternative program had a statistically significant negative effect on student outcomes (Mansell 2024).
- Students taught by alternatively certified teachers scored lower on both the 8th grade science STAAR test and the Biology end-of-course STAAR test compared to those taught by traditionally certified teachers.
- Though alternatively certified teachers comprise only 25% of the total science teacher workforce in the sample, they make up 33% of all teachers who turned over, meaning they leave at a disproportionately higher rate.
- Science teachers certified through online alternative programs have the greatest turnover rate among all alternatively certified teachers, suggesting that the least rigorous certification pathways drive the most instability in science classrooms.
- Taken together, these findings directly support our mechanism: when licensure floors are lowered and more alternatively certified teachers enter classrooms, science achievement suffers most because these teachers are both less effective and more likely to leave.

Economic and Policy Conclusions

Economic Significance

- The policy change reduced the share of licensed teachers in North Carolina charter elementary and middle schools by about 3.6 percentage points after 2014.
 - a. Results are in line with previous literature, which finds that, if given the option, schools hire non-licensed teachers (Arias & Scafidi, 2009)
 - b. While measurable, this is a modest effect size, suggesting schools adjusted hiring practices, but not dramatically.
 - c. Licensure rates also began rising again after 2018, indicating the long-run impact was limited.
- There was no statistically significant effect of decreasing teacher licensures found on student achievement, which is in line with previous literature that finds that small relaxations of teacher licensure requirements does not produce significant changes in student academic success (Backes & Goldhaber, 2023).

Policy Significance

- Relaxed teacher licensure rules appear to give charter schools more hiring flexibility that they actually do take advantage of, if only modestly.
 - a. The only modest decline in teacher licensure suggests that, even though regulations are relaxed, charter schools still work towards maintaining a relatively high level of licensed teachers
- In this sample, the modest decline in licensed teachers did not produce detectable changes in test-score achievement.
 - a. This suggests small reductions in licensure requirements may expand staffing options without immediate academic losses.

Reflection and Looking Forward

Information Needed Before Making Definitive Policy Recommendations

- Teacher quality measures: Licensure is only one proxy. Data on experience, effectiveness, retention, and classroom performance would strengthen conclusions.
- Student outcomes beyond test scores: Attendance, discipline, graduation, and long-term success
- Distributional effects: Results may differ for rural vs. urban schools, low-income students, or specific grade levels.
- Generalizability: Findings come from North Carolina and may not apply to other states with different charter systems.

Informing Potential Policy

- Learning gains were not impacted by a small change in the percentage of licensed teachers. This could inform teacher hiring policies, especially in rural areas, which may benefit from decreased hiring regulations.
- These findings also influence the larger charter school policy debate. This study found that decreased regulation of charter schools is correlated with changes in charter school action, but these changes are minimal, even four years after policy adoption.

Limitations

- Relatively small sample of charter elementary and middle schools
- Cannot isolate elementary schools only
- No data on characteristics of newly hired unlicensed teachers

Strengths

- Pre- and post-policy comparison using a difference-in-differences design.
- School fixed effects help control for time-invariant differences.
- Examines both staffing behavior and student achievement outcomes.

ADDENDUM

AI Disclosure

"During the preparation of this work, the authors used Claude and Chat GPT in order to debug coding errors and add advanced specifications in visuals and models. It was also used to generate potential academic source options for the lit review. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the presentation."

	Base	+Quad	+Level×Lic	+Charter×Lic	Full
<i>% Fully Licensed</i>	17.233***	-22.747	-17.004	68.651*	20.320
	(2.772)	(31.685)	(31.572)	(33.736)	(39.717)
<i>Charter School</i>	4.802***	4.755***	3.640**	33.854***	20.735*
	(1.268)	(1.263)	(1.277)	(6.935)	(8.333)
<i>Licensing Squared</i>		23.118	-14.238	-24.777	-29.124
		(17.992)	(18.708)	(18.999)	(21.673)
<i>Year (Centered)</i>	-0.407***	-0.422***	-0.419***	-0.432***	-0.429***
	(0.061)	(0.060)	(0.060)	(0.060)	(0.060)
<i>% White Students</i>	78.438***	78.284***	78.127***	78.461***	77.900***
	(4.969)	(4.959)	(4.924)	(4.907)	(4.913)
<i>% Black Students</i>	-9.250+	-9.127	-8.400	-7.976	-7.414
	(5.597)	(5.589)	(5.479)	(5.515)	(5.463)
<i>% Econ. Disadvantaged</i>	-165.734***	-165.752***	-165.607***	-165.660***	-165.391***
	(5.246)	(5.238)	(5.171)	(5.200)	(5.162)
<i>Num. Obs.</i>	9985	9985	9985	9985	9985
<i>R²</i>	0.683	0.683	0.689	0.685	0.690
<i>R² Adj.</i>	0.683	0.683	0.688	0.685	0.689

Note:

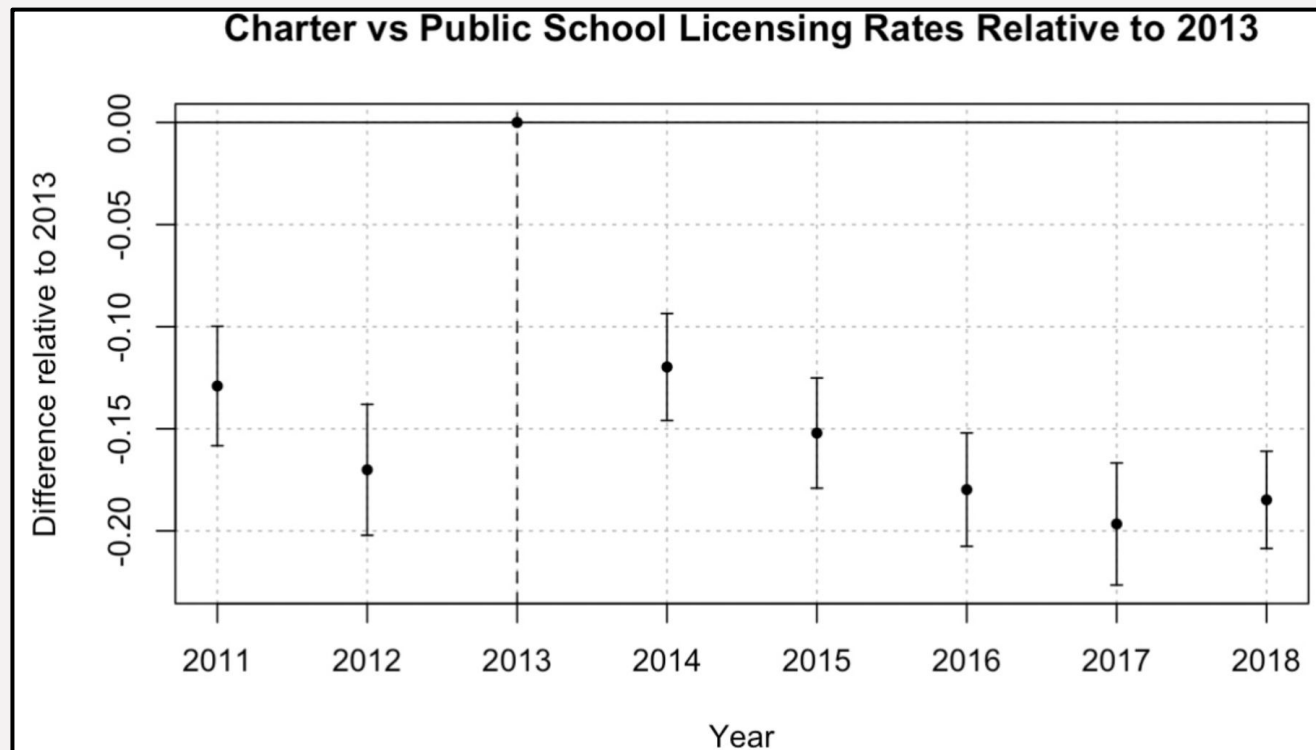
Standard errors clustered by agency_code in parentheses. Signif: * p<0.1, ** p<0.05, *** p<0.01.

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Addendum CONT.

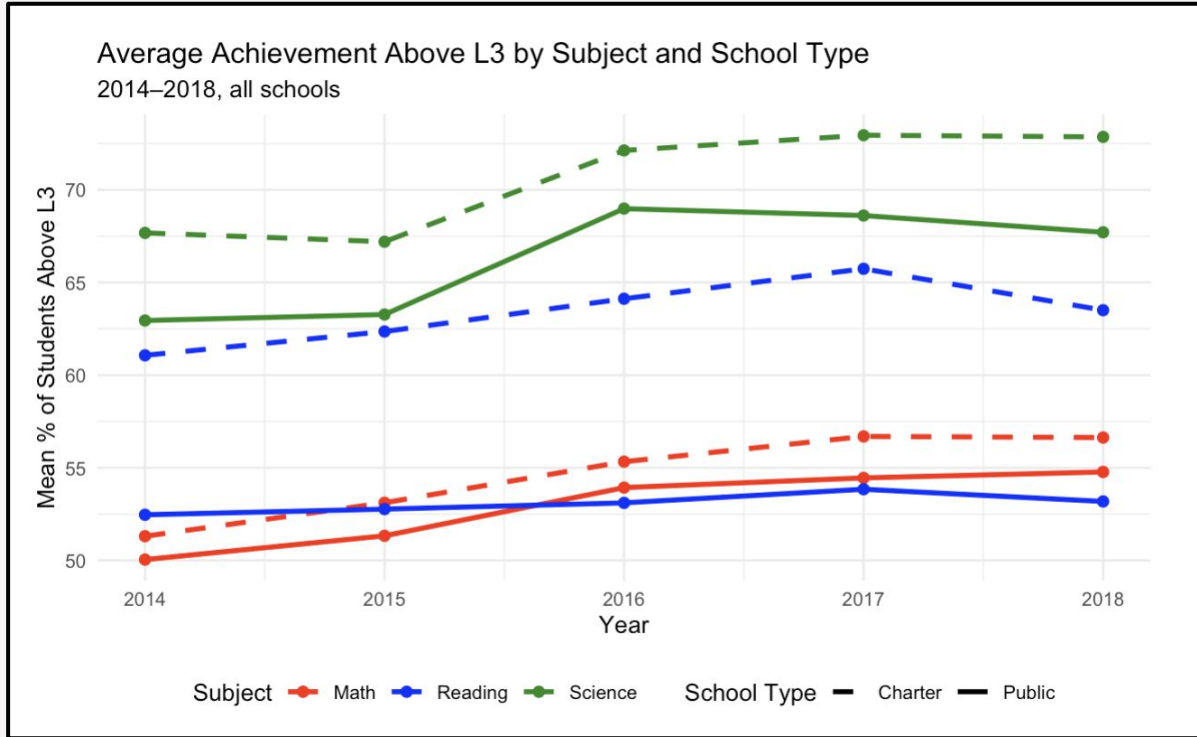
- This table replicates the main Model 2 specifications using reading pass rates as the outcome instead of math.
- In the baseline model, a 1 percentage point increase in the share of licensed teachers is associated with a 17.2 point increase in reading pass rates ($p < 0.001$).
- However, the estimate is unstable across specifications and loses statistical significance in the full model (20.3, SE = 39.7), consistent with the math results.

Addendum CONT.



This graph depicts the relationship relative to 2013 of the difference between charter and public school teacher licensing. The difference is another visual to highlight the lack of parallel trends before the policy introduction.

Addendum CONT.



This graph was created in order to understand the variation in achievement between subjects. We wanted to assess this in order to see if there are any trends we could further analyze.

If we had access to teacher turn over rate we would analyze whether that relationship was a mechanism in our study.

*Note: Science was also modeled, but the results showed the same relationship as Reading and Math.

Project Code

PDF of Knitted File

https://drive.google.com/file/d/1sQ78BeJoElj3WHv1xkRY_m3du4gGSi_C/view?usp=sharing

Raw link included in case PDF does not allow hyperlink

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